



Cambridge (CIE) IGCSE Biology



Your notes

Enzymes

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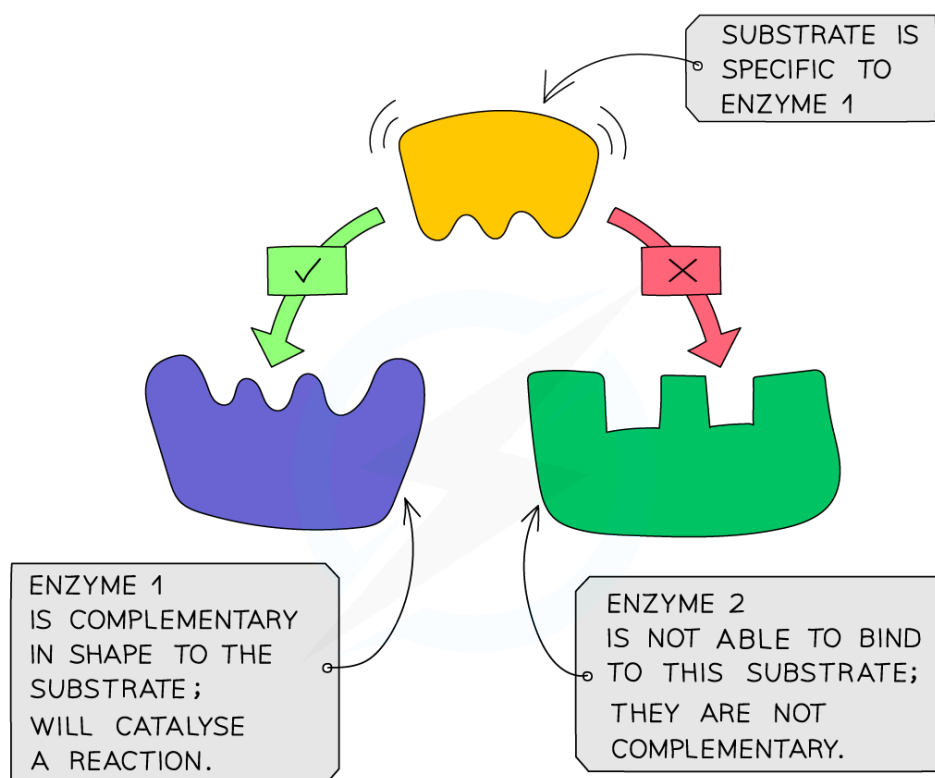
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What are enzymes?

- Enzymes are:
 - Catalysts** that **speed up** the rate of a chemical reaction **without being changed** or used up in the reaction
 - Proteins**
 - Biological catalysts** (biological because they are **made in living cells**, catalysts because they speed up the rate of chemical reactions without being changed)
 - Necessary to all living organisms as they **maintain reaction speeds** of all metabolic reactions (all the reactions that keep an organism alive) at a rate that can **sustain life**
 - For example, if we did not produce digestive enzymes, it would take around 2 – 3 weeks to digest one meal; with enzymes, it takes around 4 hours

How do enzymes work?



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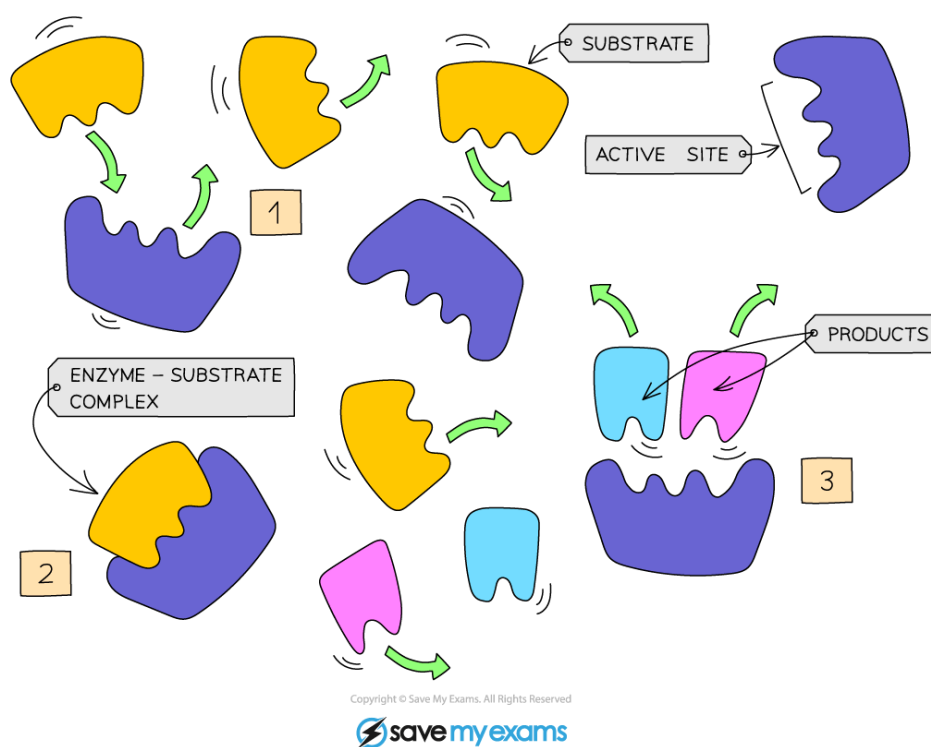


Enzyme substrate specificity

- Enzymes are **specific** to one particular substrate (molecule/s that get broken down or joined together in the reaction) as the enzyme is a complementary shape to the substrate
- The product is made from the substrate(s) and is released



Your notes



Enzyme specificity: lock and key model of enzyme activity

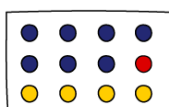
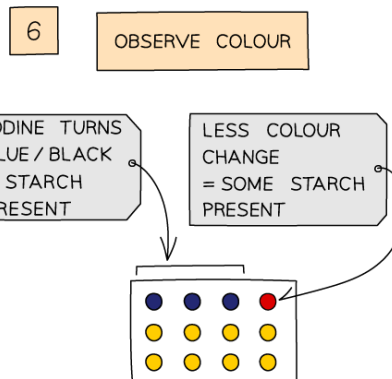
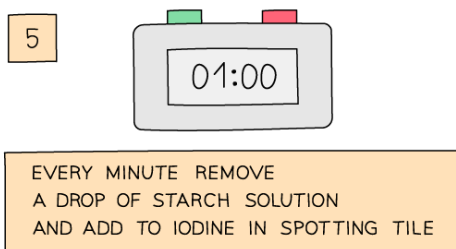
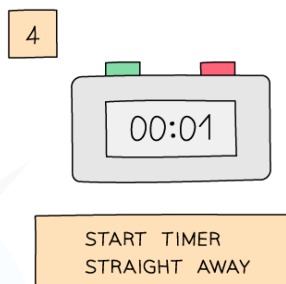
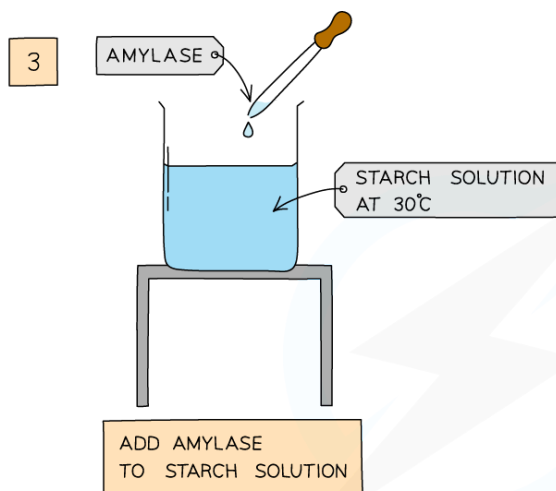
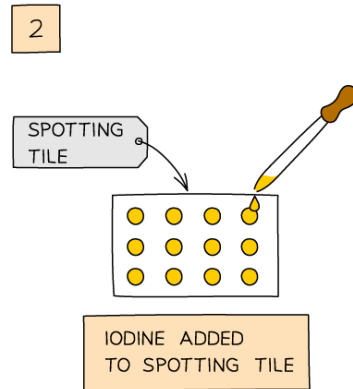
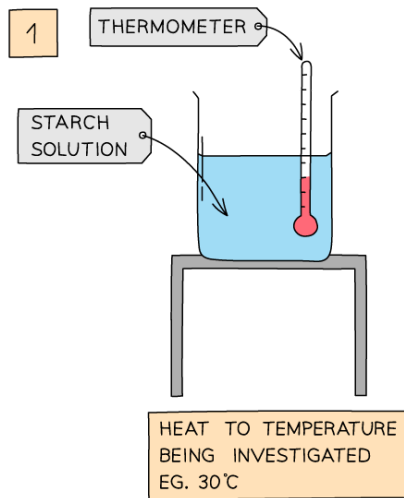


Investigating the Effect of Temperature on Amylase

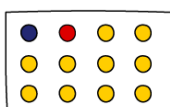
- **Starch solution** is heated to a set temperature
- **Iodine** is added to wells of a spotting tile
- **Amylase** is added to the starch solution and mixed well
- Every minute, droplets of solution are added to a new well of iodine solution
- This is continued until the iodine **stops turning blue-black** (this means there is **no more starch** left in the solution as the amylase has broken it all down)
- Time taken for the reaction to be completed is recorded
- Experiment is repeated at different temperatures
- The quicker the reaction is completed, the faster the enzyme is working



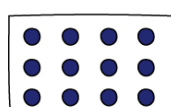
Your notes



AT 20°C
= 9 MINUTES



AT 40°C
= 3 MINUTES



AT 60°C
= NO DIGESTION
(NO COLOUR CHANGE)



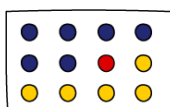
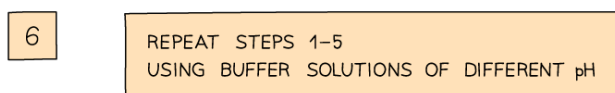
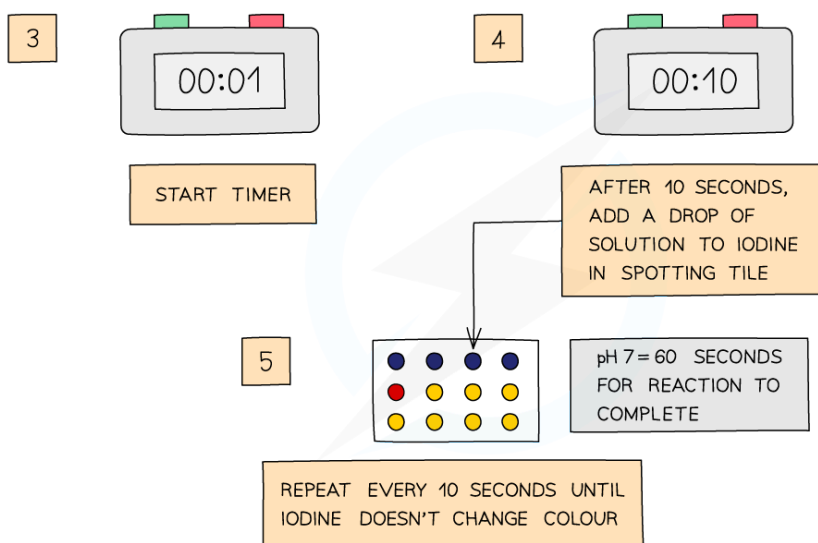
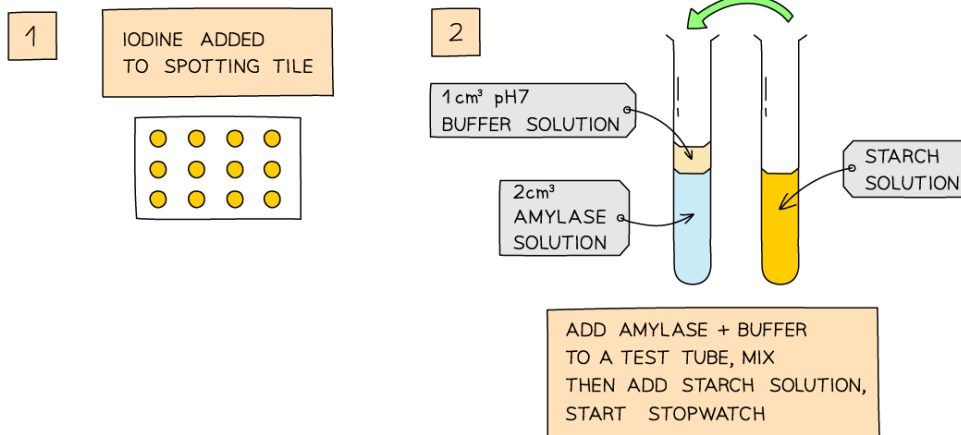
Investigating the effect of temperature on enzyme activity

Investigating the Effect of pH on Amylase

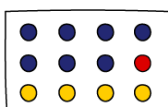
- Place single drops of **iodine** solution in rows on the tile
- Label a test tube with the pH to be tested
- Use the syringe to place 2cm^3 of **amylase** in the test tube
- Add 1cm^3 of **buffer solution** to the test tube using a syringe
- Use another test tube to add 2cm^3 of **starch solution** to the amylase and buffer solution, start the stopwatch whilst mixing using a pipette
- After 10 seconds, use a pipette to place one drop of mixture on the first drop of iodine, which should turn blue-black
- Wait another 10 seconds and place another drop of mixture on the second drop of iodine
- Repeat every 10 seconds **until iodine solution remains orange-brown**
- Repeat experiment at different pH values - the less time the iodine solution takes to remain orange-brown, the quicker all the starch has been digested and so the better the enzyme works at that pH



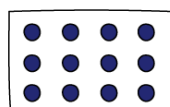
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pH 5
= 80 SECONDS



pH 9
= 90 SECONDS



pH 14
= NO REACTION

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Investigating the effect of pH on enzyme activity



Examiner Tips and Tricks

Describing and explaining experimental results for enzyme experiments is a common type of exam question so make sure you understand what is happening and, for a 7, 8 or 9, can relate this to changes in the active site of the enzyme when it has denatured, or if it is a **low temperature**, relate it to the amount of kinetic energy the molecules have.



Your notes



Enzyme Action & Specificity: Extended

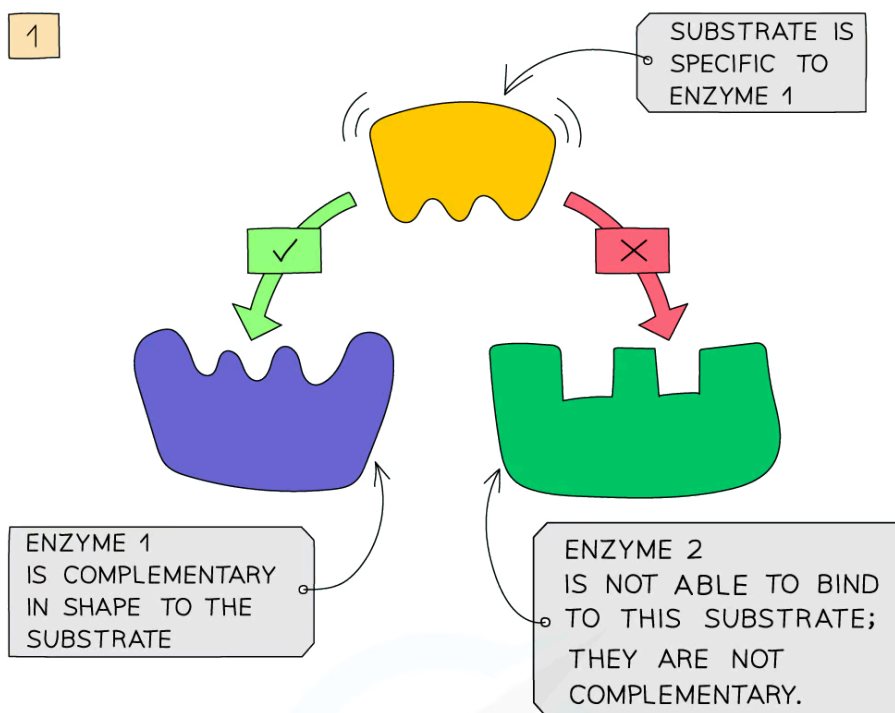
Extended Tier Only

- Enzymes are **specific** to one particular substrate(s) as the **active site** of the enzyme, where the substrate attaches, is a complementary shape to the substrate
- This is because the enzyme is a protein and has a **specific 3-D shape**
- This is known as the **lock and key hypothesis**
- When the substrate moves into the enzyme's active site they become known as the **enzyme-substrate complex**
- After the reaction has occurred, the **products** leave the enzyme's active site as they no longer fit it and it is free to take up another substrate

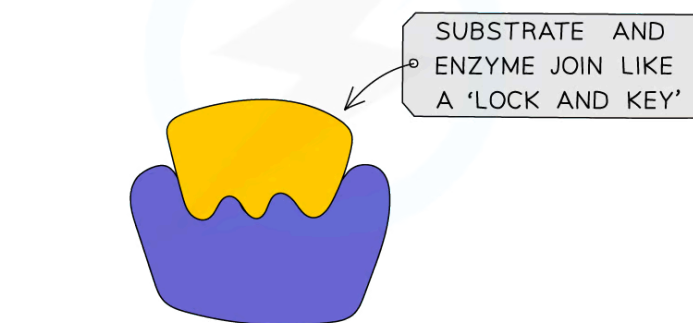


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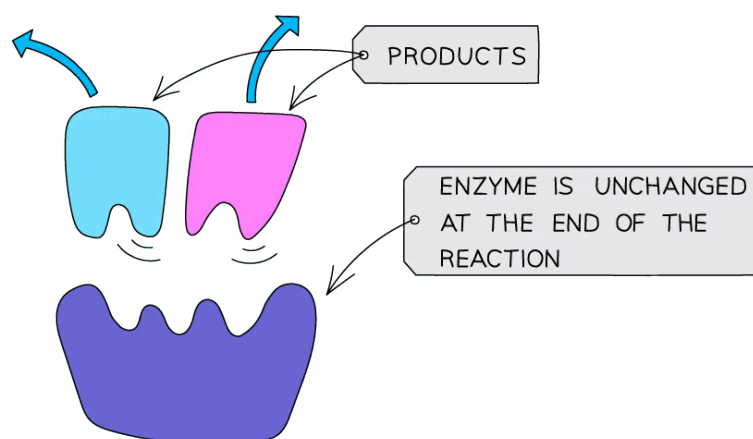
1



2



3



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How enzymes work

1. Enzymes and substrates **randomly** move about in solution
2. When an enzyme and its complementary substrate randomly **collide** - with the substrate fitting into the active site of the enzyme - an **enzyme-substrate complex** forms, and the reaction occurs.
3. A product (or products) forms from the substrate(s) which are then released from the active site. The enzyme is unchanged and will go on to catalyse further reactions.



Your notes



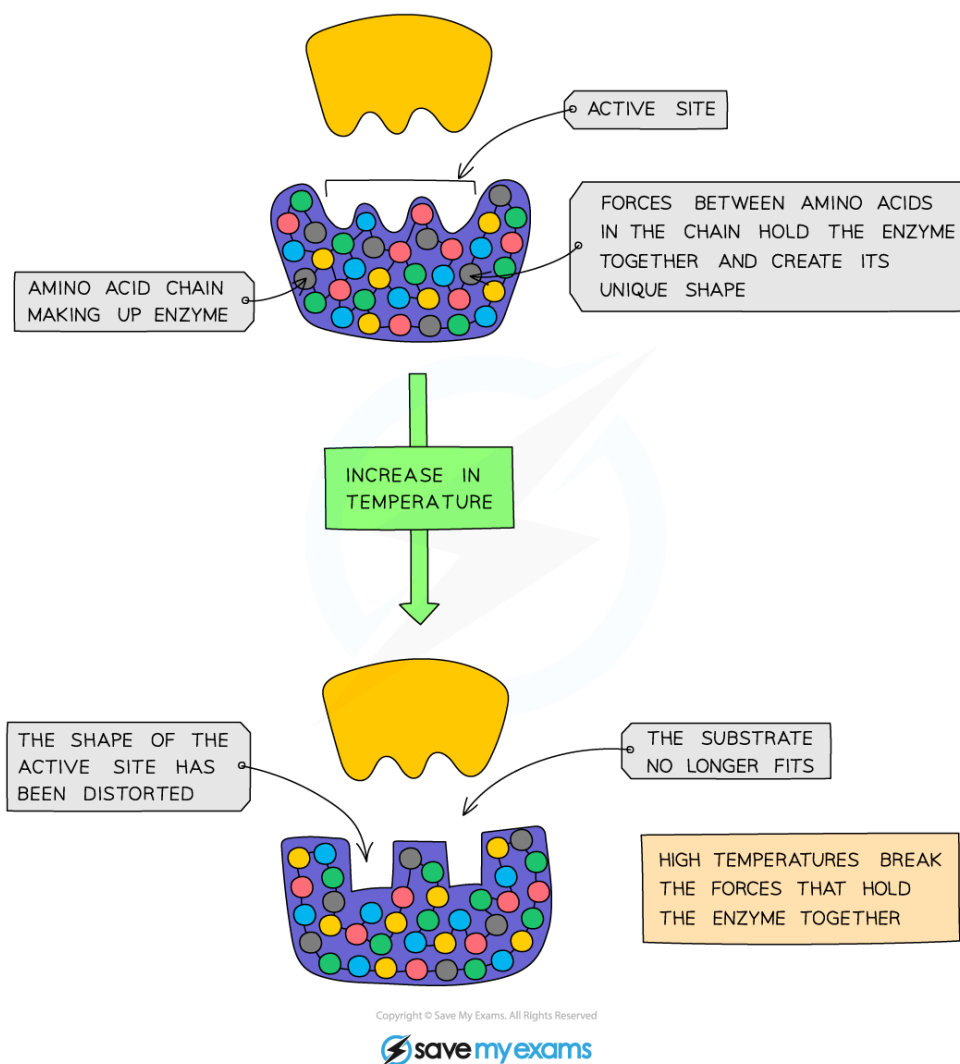
Enzymes & Temperature: Extended

Extended Tier Only

- Enzymes are **proteins** and have a **specific shape**, held in place by **bonds**
- This is extremely important around the **active site** area as the specific shape is what ensures the **substrate will fit into the active site** and enable the reaction to proceed
- Enzymes work fastest at their '**optimum temperature**' – in the human body, the optimum temperature is 37°C
- Heating to high temperatures (beyond the optimum) will **break the bonds** that hold the enzyme together and it will lose its shape – this is known as **denaturation**
- Substrates cannot fit into denatured enzymes as the shape of their active site has been lost
- Denaturation is **irreversible** – once enzymes are denatured they cannot regain their proper shape and activity will stop



Your notes

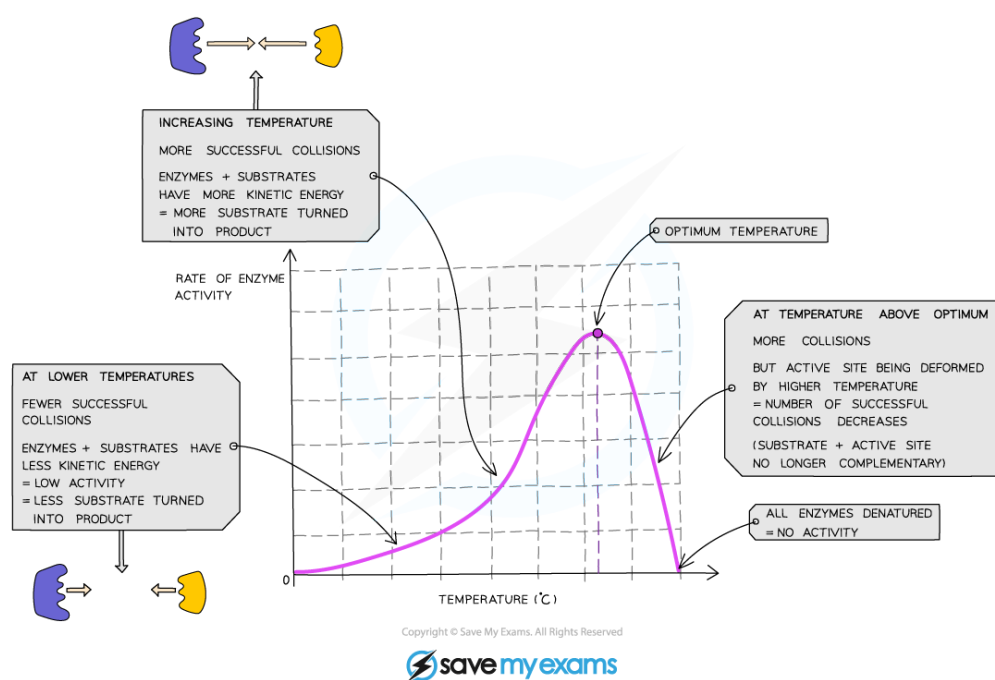


Effect of temperature on enzyme activity

- Increasing the temperature from 0°C to the optimum increases the activity of enzymes as **the more energy the molecules have the faster they move and the number of collisions with the substrate molecules increases**, leading to a faster rate of reaction
- This means that **low temperatures do not denature enzymes**, they just make them work more slowly



Your notes



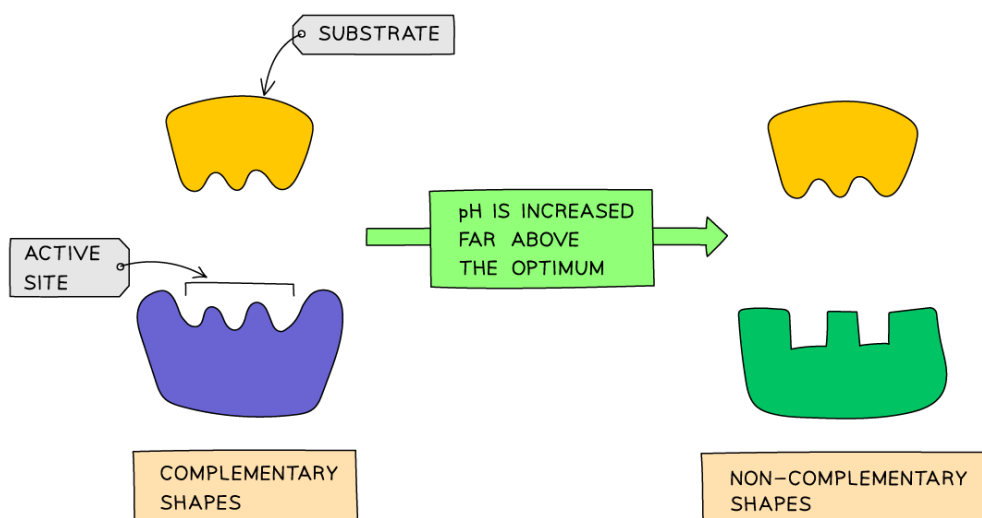
Graph showing the effect of temperature on the rate of enzyme activity



Enzymes & pH: Extended

Extended Tier Only

- The optimum pH for most enzymes is 7 but some that are produced in acidic conditions, such as the stomach, have a lower optimum pH (pH 2) and some that are produced in alkaline conditions, such as the duodenum, have a higher optimum pH (pH 8 or 9)
- If the pH is too high or too low, the bonds that hold the amino acid chain together to make up the protein can be destroyed
- This will change the shape of the active site, so the substrate can no longer fit into it, reducing the rate of activity
- Moving too far away from the optimum pH will cause the enzyme to denature and activity will stop



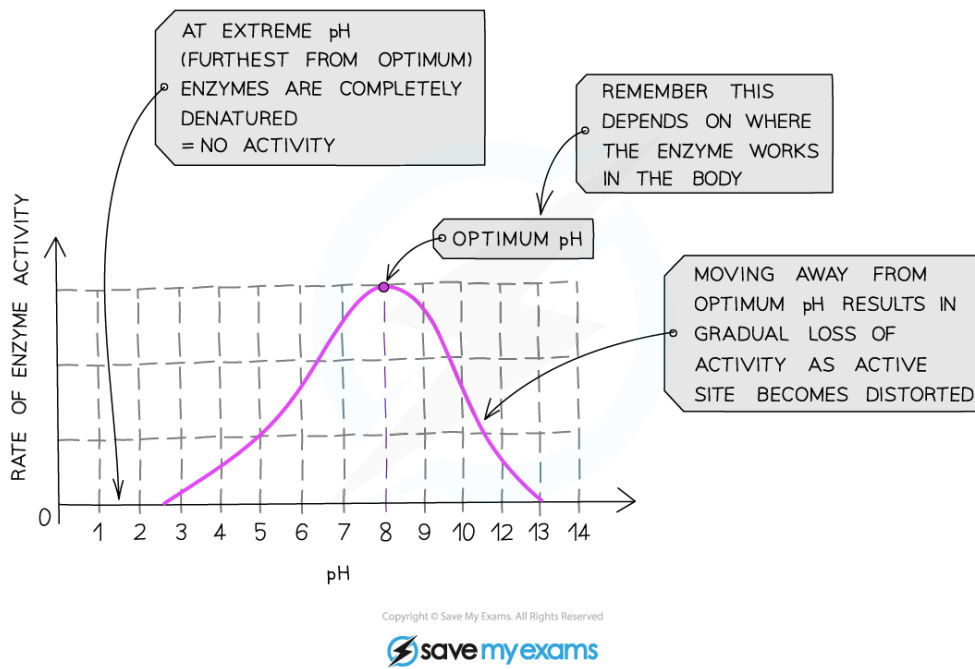
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Effect of pH on enzyme activity



Your notes



Graph showing the effect of pH on rate of activity for an enzyme from the duodenum